

# How Do Vectors Add?

## SYNOPSIS

Displacement and velocity are vector "counterparts" of distance and speed — they carry a direction as well as a size. Most quantities you'll meet are one or the other, vector or scalar, with no matching partner. Scalars add the way ordinary numbers do. Vectors don't: a vector is an arrow, and arrows add by *joining the head of one to the tail of the next* — the head-to-tail rule. A negative sign flips a vector's head and tail, pointing it the opposite way, which is how vector subtraction works. In this note we learn to add and subtract vectors diagrammatically; multiplication comes later.

## PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Distinguish scalar addition from vector addition** — scalars add like ordinary numbers; vectors add head-to-tail.
- PE2 Apply the head-to-tail rule** to join two or more vectors, keeping each vector's own magnitude and orientation.
- PE3 Recognize that a negative sign flips a vector's head and tail** — it reverses direction without changing length. That's how subtraction works.
- PE4 Draw the resultant vector** — from the tail of the first vector to the head of the last — for two or more vectors joined head-to-tail.
- PE5 Determine a resultant's magnitude and direction by measurement** (ruler and protractor) for vectors joined at any angle.
- PE6 Recognize when vectors must be rejoined** — two vectors drawn joined at the tail (or head), or separated in space, must be moved head-to-tail before you can add them.
- PE7 Relate the angle at the joint to the resultant's length** — straight ahead gives the longest resultant, doubled back the shortest, and every angle between gives something in between.



## You already know this

Head-to-tail addition isn't new — you've been doing it without the vocabulary:



### Turn-by-turn directions

"3 blocks east, then 2 blocks north." Each leg starts where the last one ended — that's head-to-tail — and where you end up is the resultant.



### Game-controller movement

Press right, then up. Your character's net move is the head-to-tail sum of the two presses, even though you made them one at a time.



### Walking it back

Walk 5 blocks east, then 5 blocks west — the second leg is the *negative* of the first. You end up exactly where you started.

## In this note, we work through

- I. What's the difference between a scalar quantity and a vector quantity?
- II. How do vectors add — and why can't we just add their lengths like numbers?
- II. What is the head-to-tail rule, and what does a negative sign do to a vector?
- IV. How do we find a resultant vector's magnitude and direction?
- V. How do we add vectors that are drawn joined at the tails — or separated in space?
- /I. How does the angle where two vectors join change the resultant's length?
- II. Which pairs of vectors produce identical resultants?



arrows join head to tail

## Play with it: the head-to-tail lab

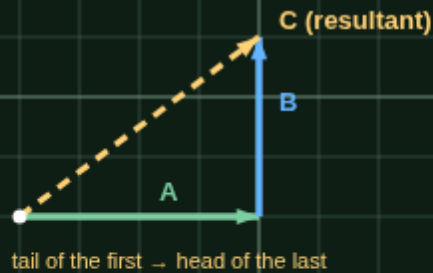
**A** and **B** start out **separated in space** — the way force vectors are usually drawn on a diagram. They can't be added as they sit there. Press **Join head-to-tail** to slide the second one across (watch it — it never rotates) until its tail meets the first one's head, then press **Show resultant**.

Three things to try. Switch to **A – B** and watch the negative sign flip B right around — a ghost of the old +B stays behind so you can see the flip. Press **Swap order** to draw B first instead: the resultant lands in

exactly the same place, every time. And swing a direction slider while watching the **joint angle** chip — that angle is what decides how long the resultant comes out.

### ► INTERACTIVE · HEAD-TO-TAIL VECTOR ADDITION

joined head-to-tail ✓ (drawing A first)



|                                                                                                                       |       |                  |     |
|-----------------------------------------------------------------------------------------------------------------------|-------|------------------|-----|
| Vector A — length                                                                                                     | 4.0 m | Vector A — angle | 0°  |
| Vector B — length                                                                                                     | 3.0 m | Vector B — angle | 90° |
| <div><div> A  = 4.0</div><div> B  = 3.0</div><div>joint angle = 90°</div><div>angle between A and B = 90°</div></div> |       |                  |     |
| <div><div>Resultant length ≈ 5.0 m</div><div>Resultant angle ≈ 37°</div></div>                                        |       |                  |     |

**This lab is interactive.** The sliders and buttons only work on screen; the figure above is one state of it, with the settings shown beneath. Open the note online to run it yourself: [https://byungmooncho.github.io/PHY151/Module\\_02/M2\\_N1\\_How\\_Do\\_Vectors\\_Add.html](https://byungmooncho.github.io/PHY151/Module_02/M2_N1_How_Do_Vectors_Add.html)

**Predict first:** two vectors, 3 m and 4 m long, are joined head-to-tail at some angle. Which of these magnitudes is *impossible* for the resultant?

7 m

5 m

1 m

8 m

### Watch: working through the note

**WALKTHROUGH VIDEO • ~10 MIN**

Watch the walkthrough in the online note.

### The joint angle at a glance

When two vectors of lengths  $a$  and  $b$  connect head-to-tail, the angle at the joint is what sets the resultant's length. Two different angles get quoted for the same picture, so the table lists both — read the column that matches the question you're answering:

| HOW THEY JOIN                                                                                                                                                                                                                                                                                                                          | JOINT ANGLE<br>at the corner | ANGLE BETWEEN<br>A and B | RESULTANT LENGTH                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------|-------------------------------------------------------|
| <b>Straight ahead</b> — same direction                                                                                                                                                                                                                                                                                                 | 180°                         | 0°                       | <b>a + b</b> — the longest possible                   |
| <b>At a corner</b> — perpendicular                                                                                                                                                                                                                                                                                                     | 90°                          | 90°                      | the diagonal of the corner — between the two extremes |
| <b>Doubled back</b> — opposite directions                                                                                                                                                                                                                                                                                              | 0°                           | 180°                     | <b>a – b</b> — the shortest possible                  |
| <b>Doubled back, equal lengths</b>                                                                                                                                                                                                                                                                                                     | 0°                           | 180°                     | <b>zero</b> — you end where you started               |
| <p>The <b>joint angle</b> is the one you measure on the diagram, at the corner where the arrows meet — that's the convention used in our guided notes and in the lab above. The <b>angle between A and B</b> is the one most textbooks and WebAssign quote; it's the supplement of the joint angle, so the two always add to 180°.</p> |                              |                          |                                                       |

**Looking ahead:** here we add vectors by *drawing* — ruler, protractor, grid paper. In Note 2 we run the rule in reverse and *resolve* one vector into a horizontal and a vertical piece, and in Note 2a trigonometry turns those drawings into calculations. Master the picture first; the math is just the picture with numbers.

• • • • • tail of the first → head of the last

### Three rules to remember



#### Head-to-tail

Join the head of one vector to the tail of the next — rejoining them first if they're drawn tail-to-tail or apart, and never rotating them.



#### C: first tail → last head

The resultant starts at the tail of the first vector and ends at the head of the last — whichever order you draw them in.



#### –B flips B

A negative sign swaps a vector's head and tail; its length never changes. Subtraction is just addition of the flipped vector.

## Exit ticket

Four quick questions — check yourself before you close this note.

**Q1. A vector quantity is best described as one with...**

- ☐ Magnitude only.
- ☐ Direction only.
- ☐ Both magnitude and direction.
- ☐ Neither — it's just a number.

**Q2. To add two vectors, you must join them...**

- ☐ Tail-to-tail.
- ☐ Head-to-head.
- ☐ Head-to-tail — the resultant runs from the tail of the first to the head of the last.
- ☐ Side by side, without moving either one.

**Q3. Two vectors, 5 m and 3 m, point in exactly opposite directions ( $180^\circ$  apart) and are joined head-to-tail. What is the magnitude of the resultant?**

m

**Q4. Two force vectors are drawn joined at their tails. Before you can add them, you must...**

- ☐ Add them exactly as drawn.
- ☐ Flip one of them.
- ☐ Separate them and rejoin them head-to-tail, keeping each vector's orientation.
- ☐ Measure the angle between them first.

**Your notes**

**Your notes**

**Guided notes — blank (PDF):**

[https://byungmooncho.github.io/PHY151/handouts/M2\\_N1\\_How\\_Do\\_Vectors\\_Add\\_Handout.pdf](https://byungmooncho.github.io/PHY151/handouts/M2_N1_How_Do_Vectors_Add_Handout.pdf)

**Completed notes (PDF):** released when Module 2 closes.